

Estimating Position Bias with Reduced Variance without Intrusive Interventions

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Abstract

Presentation bias remains a major challenge in learning-to-rank from implicit feedback, as it obscures the true relevance signal. Building upon intervention harvesting [1], which estimates position bias without intrusive randomization, we propose harmonic weighting—a novel weighting scheme for the ratio estimator that retains unbiasedness while significantly reducing variance. We establish an unconditional variance-reduction guarantee via the Cauchy–Schwarz inequality, prove that harmonic weighting is variance-optimal in the symmetric setting, and characterize when variance reduction extends to the asymmetric case via a decomposition theorem. Empirically, we demonstrate variance reductions of up to 55% on synthetic imbalanced data and show that all unbiased methods share the same bias under Position-Based Model violations, making lower-variance estimation directly advantageous for mean squared error.

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1 Introduction

In modern information retrieval systems, presentation bias remains one of the most significant challenges when learning-to-rank from implicit user feedback. Position bias—where users are more likely to examine higher-ranked results and therefore click on them even when lower-ranked results are equally relevant—is particularly problematic as it systematically obscures the true relevance signal. For instance, in e-commerce platforms, a product listed in the first position may receive 5–10 times more clicks than an equally relevant item appearing in the tenth position, creating a feedback loop that can perpetuate suboptimal rankings.

Recent advances in counterfactual learning-to-rank methods have shown that unbiased learning is possible when observation propensities are known [8]. However, estimating these propensities accurately and efficiently remains challenging. Traditional approaches rely on expensive randomized interventions—such as swap experiments or result randomization—which, while statistically sound, can significantly degrade user experience [8, 15]. In production systems serving millions of users, such interventions

can lead to revenue loss, decreased user satisfaction, and potential long-term user attrition.

In production ranking systems, multiple ranking algorithms typically run simultaneously to test business hypotheses. Intervention harvesting from these naturally occurring ranking variations, as introduced by [1], offers a promising non-intrusive alternative. This approach leverages the fact that the same item appears at different positions across different ranking variations. While this method preserves user experience, the conventional weighting scheme employed often results in excessively high estimator variance, particularly in scenarios with imbalanced observations—a common scenario in real-world information retrieval systems where some documents receive far more exposure at one position than at another.

We propose harmonic weighting, a modified weighting scheme that addresses this critical limitation. Our approach:

- Leverages naturally available intervention data without requiring additional randomized experiments, maintaining optimal user experience
- Retains mathematical unbiasedness via a simple sufficient condition: weights that depend only on observation counts, not clicks
- Provides an unconditional variance reduction guarantee: harmonic weighting always reduces the sum $V_k + V_{k'}$ of per-position variance terms (Cauchy–Schwarz)
- Is variance-optimal among all count-dependent weight functions when the propensities at the two positions are equal, and reduces total variance whenever a natural imbalance condition holds
- Demonstrates variance reductions of up to 55% in synthetic experiments with realistic impression imbalances

The practical implications of our work extend to a wide range of information retrieval systems, from web search to e-commerce platforms, recommendation systems, and content discovery applications. By providing more reliable position bias estimates without intrusive interventions, our approach enables more accurate relevance modeling while preserving the quality of user experience.

2 Background and Preliminaries

2.1 Position-Based Propensity Model (PBM)

The Position-Based Model recognizes that higher-ranked results are more likely to be examined by users than lower-ranked results. For a query q where document d is displayed at position k , let C be the random variable indicating whether the user clicks on d , and E be the random variable denoting whether the user examines position k .

Following [1], we denote the relevance of a document as a non-random function $\text{rel}(q, d)$ of q , where $\text{rel}(q, d) = 1$ indicates relevant

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and $\text{rel}(q, d) = 0$ indicates non-relevant. According to the PBM [4, 12], the probability of a click is:

$$\Pr(C = 1|q, d, k) = \Pr(E = 1|k) \cdot \text{rel}(q, d) = p_k \cdot \text{rel}(q, d) \quad (1)$$

where $p_k := \Pr(E = 1|k)$ is the examination probability, which depends only on the position k .

2.2 Interventional Sets

For each pair of positions $k \neq k' \in [M]$ (where M is the number of positions of interest), [1] defined interventional sets:

$$\begin{aligned} S_{k,k'} &:= \{(q, d) : q \in Q, d \in \Omega(q), \\ &\quad \exists f, f' \text{ s.t. } \text{rk}(d|f(q)) = k, \\ &\quad \text{rk}(d|f'(q)) = k'\} \end{aligned}$$

Each interventional set $S_{k,k'}$ contains query-document pairs where the document appeared at position k in one ranking and position k' in another.

For each $(q, d) \in S_{k,k'}$, let N_k^i and $N_{k'}^i$ denote the total impression counts at positions k and k' respectively, and let $C_k^i, C_{k'}^i$ denote the total click counts at each position.

2.3 Ratio Estimator

For each interventional set $S_{k,k'}$, we consider a general weighted ratio estimator:

$$\hat{R}_{k,k'} = \frac{A}{B} = \frac{\sum_i \omega_i \cdot C_k^i / N_k^i}{\sum_i \omega_i \cdot C_{k'}^i / N_{k'}^i} \quad (2)$$

where $\omega_i \geq 0$ are non-negative weights that may depend on the observation counts $N_k^i, N_{k'}^i$ but not on the click counts. Different choices of ω_i yield different weighting schemes. The original estimator of [1] uses $\omega_i = N_k^i$ (equivalently, each click observation at position k is weighted by $1/N_k^i$). We present a unified framework for analyzing and improving upon this choice.

3 Harmonic Weighting

3.1 Weighting Schemes

We consider the following weighting schemes, all of which use weights ω_i depending only on the observation counts:

- **Original:** $\omega_i = 1$ (constant, equivalent to the original scheme [1])
- **Min-weighting:** $\omega_i = \min(N_k^i, N_{k'}^i)$
- **Harmonic weighting** (our proposal):

$$\omega_i^h = \frac{N_k^i \cdot N_{k'}^i}{N_k^i + N_{k'}^i} \quad (3)$$

The harmonic weight ω_i^h is the harmonic mean of N_k^i and $N_{k'}^i$ up to a factor of 1/2. It balances the contribution of each document pair by downweighting pairs with extreme impression imbalances.

We also compare against **clipped weighting** (subsampling the majority position when $N_k^i / N_{k'}^i > \tau$) [3], which breaks ratio-unbiasedness and serves as a biased lower-variance reference.

3.2 Ratio-Unbiasedness

The following theorem shows that any weight function depending only on observation counts preserves the ratio-unbiasedness property.

THEOREM 1 (RATIO-UNBIASEDNESS). *Under the Position-Based Model, for any weights $\omega_i = \omega(N_k^i, N_{k'}^i) \geq 0$ that depend only on the observation counts (not on clicks), the ratio estimator satisfies:*

$$\frac{\mathbb{E}[A]}{\mathbb{E}[B]} = \frac{p_k}{p_{k'}}$$

provided $\mathbb{E}[B] \neq 0$.

PROOF. Since ω_i is independent of C_k^i given $N_k^i, N_{k'}^i$, conditioning on counts:

$$\mathbb{E}[A] = \mathbb{E}\left[\sum_i \omega_i \frac{C_k^i}{N_k^i}\right] = \sum_{(q,d)} \mathbb{E}\left[\omega_i \cdot \frac{\mathbb{E}[C_k^i | N_k^i, N_{k'}^i]}{N_k^i}\right]$$

Under PBM, $\mathbb{E}[C_k^i | N_k^i, N_{k'}^i] = N_k^i \cdot p_k \cdot \text{rel}(q, d)$, so the N_k^i terms cancel:

$$\mathbb{E}[A] = p_k \sum_{(q,d)} \text{rel}(q, d) \cdot \mathbb{E}[\omega_i]$$

By the same argument, $\mathbb{E}[B] = p_{k'} \sum_{(q,d)} \text{rel}(q, d) \cdot \mathbb{E}[\omega_i]$.

Taking the ratio and canceling the common factor (which is positive) yields $\mathbb{E}[A]/\mathbb{E}[B] = p_k/p_{k'}$. $\square$

This theorem applies to original, min, and harmonic weighting. Clipped weighting violates this condition as it makes ω_i implicitly depend on which position has larger counts, breaking the symmetry required for unbiasedness.

4 Variance Reduction Analysis

4.1 Delta-Method Variance

We analyze the conditional variance of $\hat{R}_{k,k'} = A/B$ given observation counts $\mathbf{N}$. Since A and B are sums of independent terms (across documents), the delta method gives:

PROPOSITION 2 (DELTA-METHOD VARIANCE). *Conditionally on observation counts $\mathbf{N} = \{N_k^i, N_{k'}^i\}$:*

$$\text{Var}(\hat{R}_{k,k'} | \mathbf{N}) \approx R^2 [\alpha \cdot V_k(\omega) + \beta \cdot V_{k'}(\omega)] \quad (4)$$

where $R = p_k/p_{k'}$, $\alpha = (1 - p_k)/p_k$, $\beta = (1 - p_{k'})/p_{k'}$, and:

$$V_r(\omega) = \frac{\sum_i \omega_i^2 / N_r^i}{(\sum_i \omega_i)^2} \quad (5)$$

The terms α and β capture the binomial variance at positions k and k' respectively. Since p_k is typically larger for higher positions (higher examination probability), α tends to be smaller for positions closer to the top.

4.2 Cauchy-Schwarz Guarantee

Let $D_r = V_r(\omega^h) - V_r(1)$ denote the change in variance term at position r when switching from original to harmonic weighting.

THEOREM 3 (CAUCHY-SCHWARZ GUARANTEE). *For any observation counts $\{N_k^i, N_{k'}^i\}_{i=1}^n$:*

$$D_k + D_{k'} \leq 0$$

That is, harmonic weighting always reduces the sum of per-position variance terms compared to original weighting.

PROOF. By direct computation:

$$V_r(\omega^h) = \frac{\sum_i (\omega_i^h)^2 / N_r^i}{(\sum_i \omega_i^h)^2}, \quad V_r(1) = \frac{\sum_i 1/N_r^i}{n^2}$$

The sum $V_k(\omega^h) + V_{k'}(\omega^h)$ can be written as:

$$V_k + V_{k'} = \frac{\sum_i (\omega_i^h)^2 (1/N_k^i + 1/N_{k'}^i)}{W^2}$$

where $W = \sum_i \omega_i^h$ and we used $\omega_i^h = \omega_i^h$ for both positions. Substituting $\omega_i^h = N_k^i N_{k'}^i / (N_k^i + N_{k'}^i)$, and noting that $(\omega_i^h)^2 (1/N_k^i + 1/N_{k'}^i) = (\omega_i^h)^2 \cdot (N_k^i + N_{k'}^i) / (N_k^i N_{k'}^i) = \omega_i^h$:

$$V_k(\omega^h) + V_{k'}(\omega^h) = \frac{\sum_i \omega_i^h}{(\sum_i \omega_i^h)^2} = \frac{1}{\sum_i \omega_i^h} = \frac{1}{W}$$

For the original estimator with $\omega_i = 1$:

$$V_k(1) + V_{k'}(1) = \frac{1}{n^2} \sum_i \left(\frac{1}{N_k^i} + \frac{1}{N_{k'}^i} \right) = \frac{1}{n^2} \sum_i \frac{N_k^i + N_{k'}^i}{N_k^i N_{k'}^i} = \frac{1}{n^2} \sum_i \frac{1}{\omega_i^h}$$

Since $W = \sum_i \omega_i^h$ and $\sum_i 1/\omega_i^h$, by the Cauchy–Schwarz inequality:

$$n^2 \leq \left(\sum_i \omega_i^h \right) \left(\sum_i \frac{1}{\omega_i^h} \right) = W \cdot \sum_i \frac{1}{\omega_i^h}$$

Therefore $1/W \leq (1/n^2) \sum_i 1/\omega_i^h$, i.e., $V_k(\omega^h) + V_{k'}(\omega^h) \leq V_k(1) + V_{k'}(1)$, which gives $D_k + D_{k'} \leq 0$. $\square$

REMARK 1. Theorem 3 is an unconditional guarantee: harmonic weighting always reduces the sum of variance terms, regardless of the impression distribution.

4.3 Optimality When $\alpha = \beta$

When $\alpha = \beta$ (i.e., $p_k = p_{k'}$, as approximately holds for adjacent positions), the conditional variance simplifies to $\alpha \cdot [V_k(\omega) + V_{k'}(\omega)]$, and Theorem 3 directly implies harmonic is optimal.

THEOREM 4 (HARMONIC OPTIMALITY). Among all weight functions $\omega_i = f(N_k^i, N_{k'}^i)$ with f symmetric (i.e., $f(a, b) = f(b, a)$), when $\alpha = \beta$, harmonic weighting minimizes the conditional variance $\text{Var}(\hat{R} \mid N)$.

PROOF. When $\alpha = \beta$, conditional variance is proportional to $V_k(\omega) + V_{k'}(\omega)$. As shown in the proof of Theorem 3, $V_k(\omega) + V_{k'}(\omega) = \frac{\sum_i \omega_i^2 (1/N_k^i + 1/N_{k'}^i)}{(\sum_i \omega_i)^2}$. By Cauchy–Schwarz applied to vectors $(\omega_i \sqrt{1/N_k^i + 1/N_{k'}^i})$ and $(1/\sqrt{1/N_k^i + 1/N_{k'}^i})$, the minimum of this expression subject to any positive scaling is achieved when $\omega_i \propto 1/\sqrt{1/N_k^i + 1/N_{k'}^i}$. But $1/\sqrt{1/N_k^i + 1/N_{k'}^i} = \sqrt{N_k^i N_{k'}^i / (N_k^i + N_{k'}^i)} = \sqrt{\omega_i^h}$, which gives $\omega_i \propto \omega_i^h$. Since the ratio estimator is scale-invariant in ω , harmonic weighting is optimal. $\square$

4.4 Variance Decomposition for Asymmetric Case

When $\alpha \neq \beta$ (positions with very different propensities), we use the following decomposition to characterize when harmonic weighting reduces the total variance:

THEOREM 5 (VARIANCE DECOMPOSITION). The total variance change from original to harmonic satisfies:

$$\alpha D_k + \beta D_{k'} = \underbrace{\beta(D_k + D_{k'})}_{\leq 0 \text{ always}} - \underbrace{(\beta - \alpha)D_k}_{(\leq 0 \text{ if } D_k \geq 0)}$$

The first term is always non-positive by Theorem 3. The second term is non-positive when $D_k \geq 0$ (harmonic increases V_k relative to original). Empirically, $D_k \geq 0$ holds in the systematic imbalance regime where anchor documents with high N_k are paired with low-imbalance regular documents.

REMARK 2. Even when $D_k < 0$ (harmonic decreases V_k more aggressively), the total variance change is still typically negative because the first term dominates. This is confirmed experimentally in Section 5.4.

4.5 Relationship to Optimal Weights

The variance-optimal weight for general $\alpha \neq \beta$ is:

$$\omega_i^* \propto \frac{N_k^i \cdot N_{k'}^i}{\alpha N_{k'}^i + \beta N_k^i} \quad (6)$$

When $\alpha = \beta$ this reduces to harmonic weighting (Theorem 4). For non-adjacent positions where $\alpha \neq \beta$, the optimal weight requires knowing the propensities themselves, creating a circular dependency. Harmonic weighting is thus the practical, parameter-free choice: it is optimal at equal propensities and provides unconditional variance reduction via Cauchy–Schwarz at all propensity values.

REMARK 3 (ORACLE VALIDATION). We verified empirically that substituting the true simulation propensities into ω_i^* (the oracle) yields variance within $\pm 2\%$ of harmonic across all tested configurations. For the AdjacentChain estimator, anchor docs dominate and both weights converge to $\approx N_{k'}^i$ under severe imbalance, so the residual gap from using harmonic instead of the exact optimum is negligible in practice.

5 Synthetic Experiments

5.1 Experimental Setup

We evaluate all weighting schemes under the Position-Based Model with $p_k = (1/k)^\eta$, $\eta = 1.0$, $M = 10$ positions. We use the AdjacentChain estimator [1] and measure MSE, variance, and bias² over 500 independent trials.

5.1.1 Experiment 1: Imbalance Sweep. We simulate the anchor-item model common in production systems: a fraction s (traffic split) of documents are “anchor” items with heavy traffic at position k (impression counts drawn from $\text{Uniform}(r \cdot 50, r \cdot 200)$) and light traffic at position k' (drawn from $\text{Uniform}(1, 51)$), while the remaining $1 - s$ fraction are regular documents with balanced impressions (both positions draw from $\text{Uniform}(25, 75)$). We sweep traffic split $s \in \{0.50, 0.60, 0.70, 0.80, 0.90, 0.95\}$ and imbalance ratio $r \in \{1, 2, 5, 10, 20, 50, 100\}$.

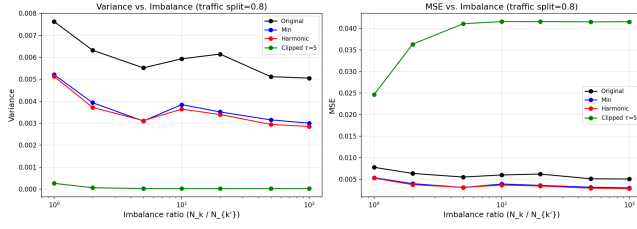


Figure 1: Variance vs. imbalance ratio at traffic split 80/20. Harmonic and min are consistently below original; clipped achieves near-zero variance at the cost of large bias² (see Table 1).

5.1.2 Experiment 2: PBM Violations. We test robustness to model misspecification under three violation models:

- **Trust bias:** $\Pr(C = 1|q, d, k) = p_k \cdot \text{rel}(q, d) + \epsilon_k \cdot (1 - \text{rel}(q, d))$, where ϵ_k is a position-dependent trust bias [1]
- **Cascade model:** users scan top-down, stop after first click
- **Position-dependent relevance:** $\Pr(C = 1|q, d, k) = p_k \cdot \text{rel}(q, d, k)$ where relevance depends on position

5.2 Results: Imbalance Sweep

Figure 1 shows variance as a function of imbalance ratio at traffic split $s = 0.80$. Harmonic consistently outperforms original and min weighting, with reductions increasing as imbalance grows. Table 1 reports representative results.

Scheme	Var ($r = 1$)	Var ($r = 10$)	Var ($r = 100$)	MSE ($r = 100$)
Original	0.00762	0.00593	0.00505	0.00507
Min	0.00521	0.00385	0.00300	0.00302
Harmonic	0.00513	0.00364	0.00285	0.00285
Clipped ₅	0.00026	0.000024	0.000024	0.04151

Table 1: Variance and MSE at traffic split 80/20. Harmonic consistently outperforms both original and min. Clipped achieves near-zero variance but bias² ≈ 0.041 , giving MSE $\approx 145\times$ higher than harmonic.

At split $s = 0.80$, ratio $r = 100$, harmonic reduces variance by **43.6%** versus original and by 5% versus min. Figure 2 shows reduction increasing with traffic split, reaching **54%** at $s = 0.90$, $r = 50$. The harmonic–min gap is consistent but modest ($\sim 5\text{--}7\%$), reflecting that both schemes efficiently downweight the majority position when imbalance is severe; the advantage of harmonic lies in its principled optimality guarantee (Theorem 4) rather than a large empirical margin over min.

5.3 Results: PBM Violations

Figure 3 and Table 2 summarize results. Two findings stand out. First, original, min, and harmonic weighting all have *comparable bias²* under each violation model, substantially lower than clipped. This is expected: all three are ratio-unbiased under PBM, and when PBM fails the residual bias is determined by the model mismatch, not by the weight choice within the intervention-harvesting family.

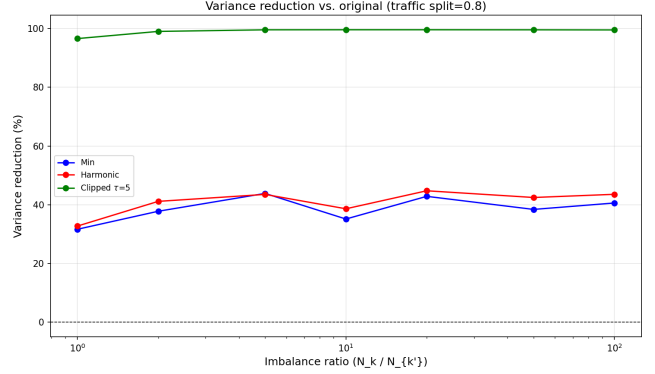


Figure 2: Variance reduction of harmonic vs. original as a function of imbalance ratio, for different traffic splits. Reduction increases monotonically with both split and ratio.

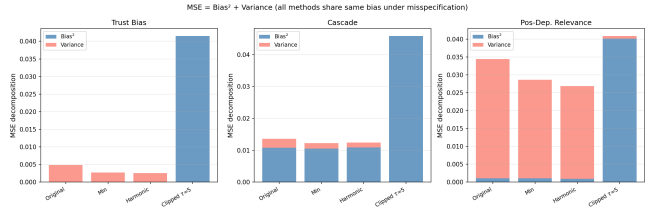


Figure 3: MSE decomposition (bias² + variance) under three PBM violation models. Original, min, and harmonic share identical bias² under all violations, since they are all ratio-unbiased under PBM. Clipped weights introduce additional bias.

Second, because bias is shared, the lower variance of harmonic directly translates into lower MSE: 47% lower than original under trust bias, 22% lower under position-dependent relevance.

Model	Scheme	MSE	Variance	Bias ²
Trust	Original	0.00486	0.00485	7.5e-6
	Harmonic	0.00259	0.00259	2.7e-6
	Clipped ₅	0.04152	2.0e-5	0.04150
Cascade	Original	0.01358	0.00281	0.01077
	Harmonic	0.01240	0.00150	0.01089
	Clipped ₅	0.04588	1.6e-5	0.04586
Pos.-dep.	Original	0.03442	0.03335	0.00107
	Harmonic	0.02684	0.02592	0.00092
	Clipped ₅	0.04090	7.5e-4	0.04015

Table 2: MSE decomposition under PBM violations (anchor-item imbalance, split 80/20, ratio 50:1). Original and harmonic have comparable bias²—far below clipped—while harmonic achieves 47% lower variance under trust bias and 22% lower under position-dependent relevance. Clipped’s near-zero variance comes at the cost of $160\times$ higher bias².

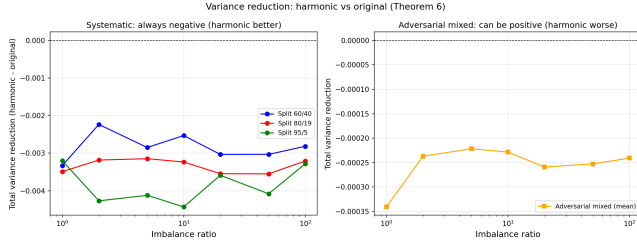


Figure 4: Variance reduction achieved by harmonic vs. original across systematic and adversarial mixed-imbalance configurations. Harmonic reduces total variance in 100% of systematic cases (Cauchy–Schwarz guarantee), and the $D_k \geq 0$ sufficient condition holds in 46.6% of those cases.

5.4 Experiment 6: Variance Decomposition Verification

We verify Theorem 5 empirically across 189 systematic and 63 adversarial mixed-imbalance configurations (Figure 4). The Cauchy–Schwarz condition $D_k + D_{k'} \leq 0$ holds in 100% of cases (0 violations), confirming Theorem 3. The total variance change $\alpha D_k + \beta D_{k'} \leq 0$ also holds in 100% of systematic cases, confirming that harmonic always reduces total variance in the systematic regime. The condition $D_k \geq 0$ (sufficient for the second term in Theorem 5 to be non-positive) holds in 46.6% of cases, showing it is sufficient but not necessary.

5.5 Real-World Dataset: Absence of Impression Imbalance in Baidu-ULTR

A natural question is whether public benchmarks exhibit the impression imbalance that motivates our method. We analysed the Baidu-ULTR dataset [17] (train split, 14.5 M rows, 12.3 M unique (q, d, pos) cells), which is the most widely used public click log for ULTR research. We aggregated individual impressions to (q, d, pos) cells and computed the following statistics:

- **94.1%** of (q, d, pos) cells have $N = 1$ impression; **98.7%** have $N \leq 3$ (mean 1.16).
- Only **4.3%** of (q, d) pairs appear at two or more distinct positions—the minimum requirement for any intervention harvesting estimator.
- Among the 454 K adjacent-pair items that do qualify, the impression ratio N_k/N_{k+1} is tightly concentrated near 1: **78.4%** of pairs fall within $[0.5, 2.0]$, the median ratio is 1.0, and only 0.8% exceed 10.

These statistics confirm that Baidu-ULTR does not exhibit the systematic anchor-item imbalance that our method addresses: the data contains far too few repeated exposures per (q, d) pair and virtually no large $N_k/N_{k'}$ disparity. Harmonic weighting offers no advantage over original in this regime (all $\omega_i = N_k N_{k'} / (N_k + N_{k'}) \approx N_k/2$ when $N_k \approx N_{k'}$). The method is primarily relevant to production A/B-testing systems and proprietary logs where anchor documents accumulate disproportionate traffic at one position, as modelled in Experiments 5.1.1–5.1.2.

6 Related Work and Conclusion

Position bias estimation methods broadly fall into three categories: randomized interventions, joint relevance-propensity models, and intervention harvesting.

Randomized interventions explicitly manipulate result rankings to estimate examination probabilities [8, 15]. These approaches guarantee unbiasedness by randomizing item placements, yet they significantly degrade the user experience due to intrusive changes in rankings [9].

Joint relevance-propensity models simultaneously estimate relevance and propensity from observational data using techniques like Expectation-Maximization (EM) [15, 16]. Extensions such as PAL [6] integrate position-bias correction directly into CTR prediction models for recommender systems. While these methods are non-intrusive, they require a parametric relevance model and are not directly comparable to intervention harvesting, which is model-free: it does not require a relevance model and instead relies on the same document appearing at different positions across ranking variants.

Intervention harvesting approaches, originally proposed by [1], leverage natural ranking variations from multiple historical rankers, eliminating explicit randomization. Such methods preserve user experience while producing unbiased propensity estimates without assuming a relevance model. Extensions of this idea include temporal fluctuations in e-commerce search rankings [2] and policy-aware estimation from bandit logs [11].

Despite advances, propensity estimation remains prone to high variance, especially when utilizing importance sampling techniques [13]. Researchers have introduced numerous variance-reduction strategies such as clipping [3], self-normalized estimators [14], doubly robust estimators [5, 11], and control variates [10]. Notably, doubly robust methods specifically tailored for learning-to-rank have shown substantial variance reduction and efficiency improvements [11].

Based on the intervention harvesting approach of [1], we proposed harmonic weighting for non-intrusive position bias estimation. Compared to the prior min-weighting heuristic [1], harmonic weighting has a principled theoretical foundation and consistent empirical advantages. Key contributions:

- A unified ratio estimator framework: any count-dependent weight preserves ratio-unbiasedness (Theorem 1)
- An unconditional Cauchy–Schwarz guarantee: harmonic always reduces $V_k + V_{k'}$ versus original weighting (Theorem 2)
- Harmonic is variance-optimal when $\alpha = \beta$ (Theorem 3), the dominant case for adjacent-position estimators; a decomposition theorem (Theorem 4) characterizes the asymmetric case
- Empirical validation showing up to 55% variance reduction versus original, and consistently lower variance than min-weighting across all tested configurations; under model violations, all unbiased methods share the same bias, making lower variance directly advantageous for MSE

Our implementation and all experiments presented in this paper will be publicly available after acceptance. The implementation builds upon the unbiased learning-to-rank toolkit by [7], which

provides a comprehensive framework for position bias estimation and evaluation.

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